



LABORATORY ACTIVITY NO. 1.4.4

Using Management Interfaces

This lab activity is designed to familiarize users with the principal management interfaces available in Windows 11. This guides students through accessing and utilizing various tools essential for system configuration, monitoring, and basic troubleshooting within the Windows 11 environment. Key areas covered include: exploring the Quick Access menu for power users, navigating the redesigned Settings app for system information and configurations, managing running processes and monitoring system resources using Task Manager, and delving into administrative tools like Computer Management (which provides access to Event Viewer, Services, and Device Manager). The lab also introduces command-line interfaces, specifically Windows PowerShell and Command Prompt, demonstrating their different capabilities for network configuration tasks.

I. OBJECTIVES

Upon completion of this laboratory exercise, students are expected to and meet the following intended learning outcomes:

Activity Objectives	Intended Learning Outcomes		
	ILO1	ILO2	ILO3
1) access and navigate key Windows 11 management interfaces, including the Quick Access menu, Settings app, Task Manager, and Control Panel.	•	•	
2) utilize management tools to perform basic system tasks, such as ending processes and Browse system information.	•	•	
3) understand and differentiate between various command-line interfaces (CLIs), including Windows PowerShell and Command Prompt, and their command capabilities.	•	•	

Where:

- ILO1 – Exhibit thorough demonstration of the concept
- ILO2 – Being able to perform the activity independently
- ILO3 – Design and evaluate emerging IT solutions

II. DISCUSSION

- **Management Interfaces:**



- **GUI Tools:** Windows provides various snap-ins (specialized management tools) accessible through the Microsoft Management Console (MMC) or standalone applications for system management.
 - **Command Line Interface (CLI):** Powerful text-based interfaces such as Windows PowerShell, Command Prompt (for Windows), and Bash (for Linux) allow for efficient OS management.
- **Computer Management Console:** This comprehensive administrative console provides a centralized point of access to several "snap-ins" for system administration, including:
 - Task Scheduler for automating tasks.
 - Event Viewer for reviewing system and application logs.
 - Shared Folders for managing network shares.
 - Local Users and Groups for managing user accounts and security groups on the local machine.
 - Performance Monitor for analyzing system performance data.
 - Device Manager for managing hardware devices and their drivers.
 - Disk Management for configuring disk drives and partitions.
 - Services for managing background processes and system services.
- **Windows Settings and Control Panel:**
 - **Settings App:** The modern interface for configuring various aspects of Windows, categorized into sections like System, Devices, Network & Internet, Personalization, Apps, Accounts, etc.
 - **Control Panel:** The traditional interface, which still contains advanced configuration options and legacy tools not yet fully integrated into the Settings app (e.g., Administrative Tools, Programs and Features).
- **Access Control and Protection:** This feature ensures that only authorized users can access computer systems, networks, and data. Users are identified via accounts and must authenticate with credentials unique to them. Accounts are assigned specific privileges or rights defining what actions they can perform.
- **Account Types and User Account Control (UAC):**
 - **Administrator Accounts:** Possess full control over the operating system, allowing for system-wide changes.
 - **Standard User Accounts:** Have limited privileges, designed to protect the system from unauthorized or accidental changes.
 - **Principle of Least Privilege:** A security best practice that advocates assigning the minimum necessary rights and permissions to user accounts.
 - **User Account Control (UAC):** A security feature that prompts for explicit administrator authorization before allowing significant system changes, even if the user is logged in as an administrator, to prevent malicious software from making unauthorized modifications.



- **Creating and Managing User Accounts:** Discussions cover the differences between cloud-based Microsoft accounts and local accounts (which are device-specific). It also mentions using the Account settings app for basic management (e.g., changing passwords) and the Local Users and Groups snap-in for more advanced local account configurations.

III. MATERIALS

A Windows 11 installed computer system.

IV. PROCEDURE

For this activity, you will execute the provided step-by-step instructions. As you complete each step, capture a screenshot of the resulting output or completed action that directly demonstrates successful execution of that instruction. Paste each screenshot into the designated area within your lab document.

Important Considerations:

- **Operating System Differences:** Considering the students familiarity with the Windows 10 environment, the core functionality of many management tools remains consistent, the user interface (UI) and navigation paths within Windows 11 have been updated. Users should be prepared for slight variations in menu layouts (e.g., the Start Menu, Settings app, and context menus).
- **Administrator Privileges:** Many of the tasks and interfaces explored in this lab, particularly those within the Quick Access menu and Computer Management, require **administrator privileges** to access and modify settings effectively. Ensure you are logged in with an account that has administrative rights. In our case we will be using a virtual machine.
- **Risk of Data Loss:** A critical warning remains: when using Task Manager to "End task" for an application that is processing unsaved data, **the information will be lost**. Always exercise extreme caution when terminating processes, especially if you have unsaved work.
- **Command-Line Interface Nuances:** While both Windows PowerShell and Command Prompt are valuable command-line interfaces, they have different command sets. Windows PowerShell is more powerful and can understand most older Windows commands (like `ipconfig`), but the Command Prompt environment **cannot understand PowerShell-specific commands** (like `Get-NetIPConfiguration`), which will result in an error message. Be mindful of which environment you are using for specific commands.
- **Impact of System Changes:** Exploring and modifying settings within management interfaces such as Disk Management, Services, or Device Manager can significantly impact your system's stability and performance. It is always advisable to proceed with caution and a clear understanding of the potential consequences of any changes you make.



Members:

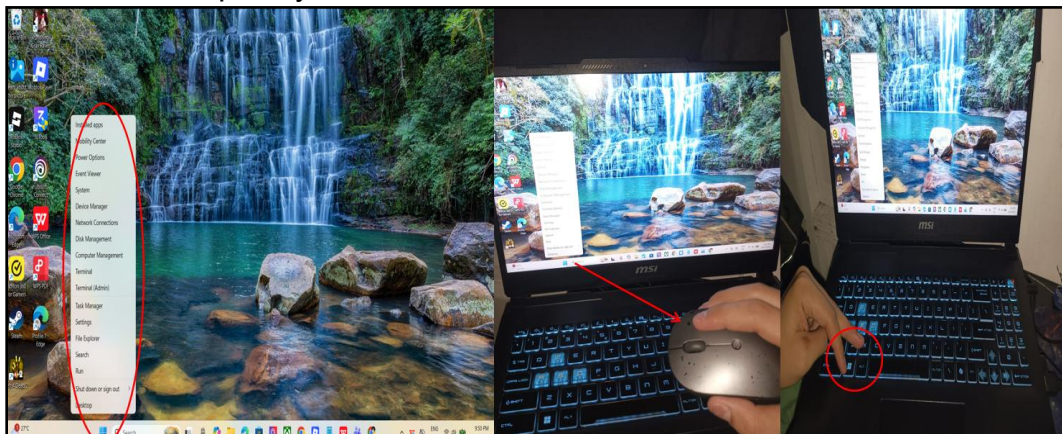
1. Francisco, Sean Gabriel
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Steps:

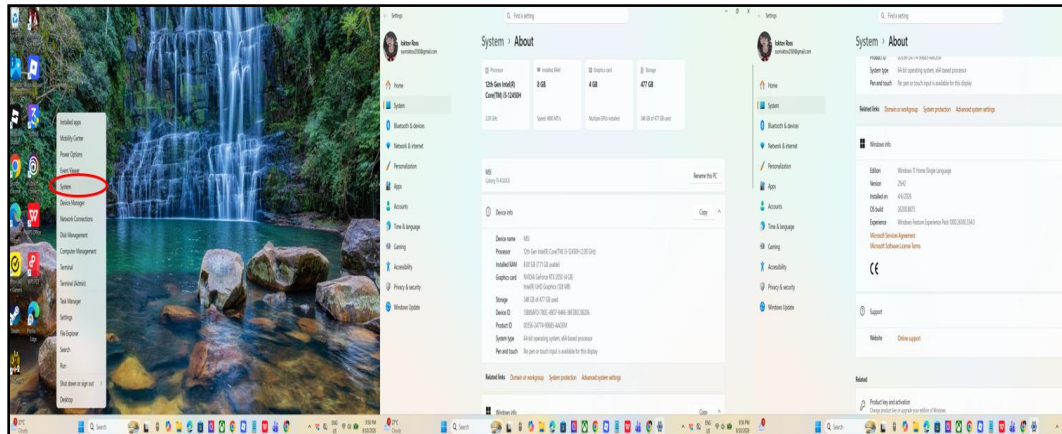
1. **Start your computer and sign in to Windows 11.**



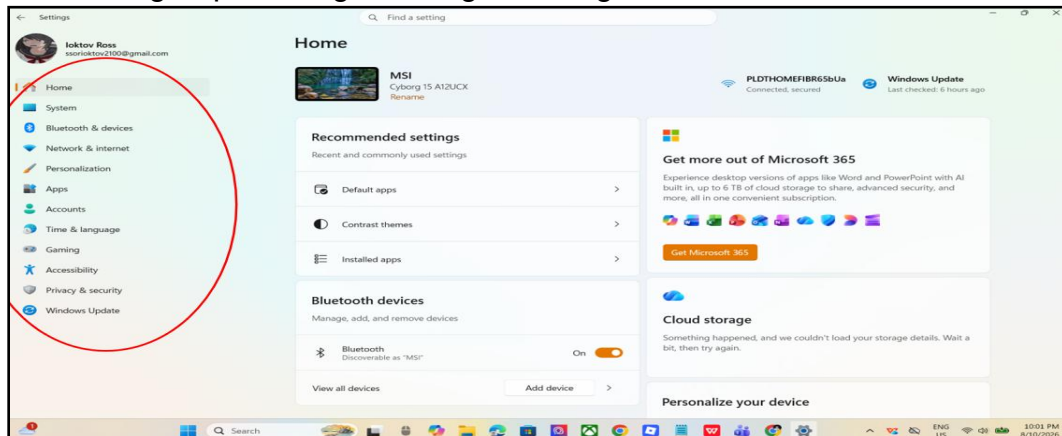
2. **Access the Quick Access Menu:** Right-click the **Start** button (typically centered on the Taskbar) or press Windows key + X. This action opens the Quick Access menu (also known as the Power User Menu), which provides shortcuts to frequently used administrative tools.



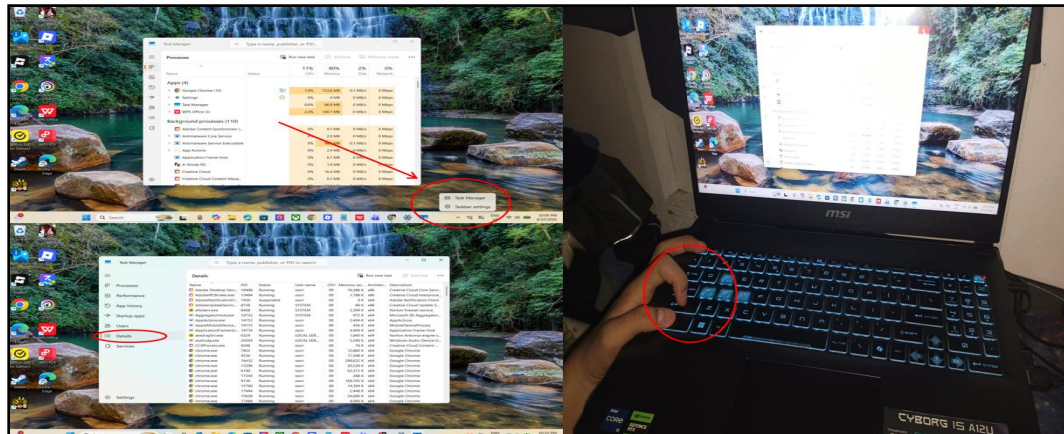
3. **View System Information:** From the Quick Access menu, select **System**. This will open the **Settings** app to the System page, providing an overview of your PC's specifications and the installed Windows 11 version.



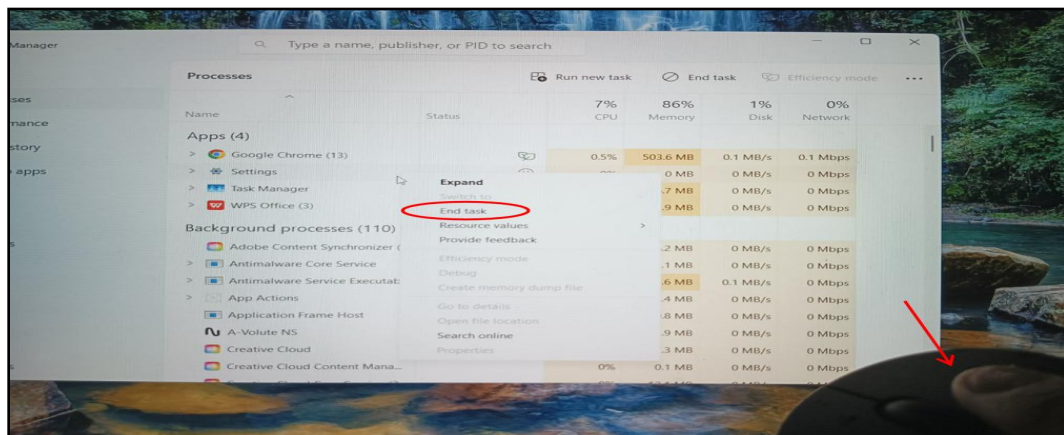
4. **Explore Settings App:** Take some time to browse other categories and pages within the **Settings** app. Notice the redesigned navigation pane on the left, which groups settings into logical categories.



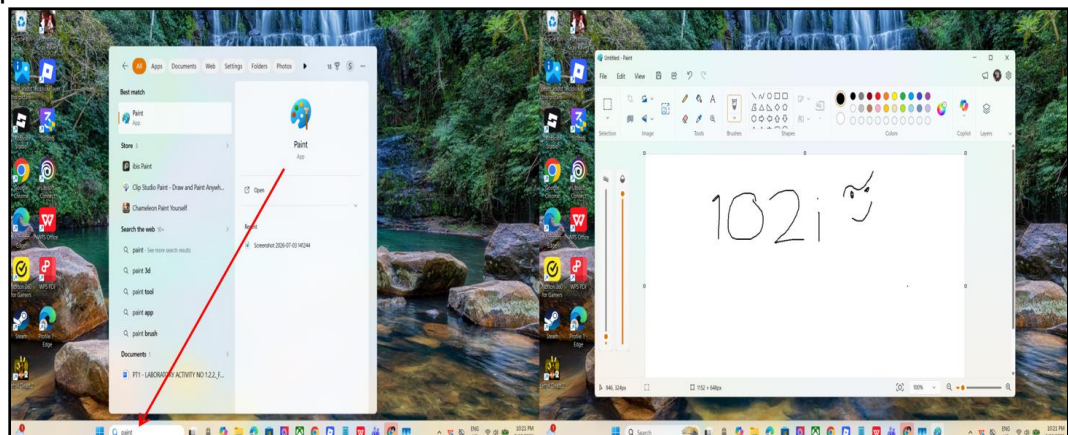
5. **Open Task Manager:** Press CTRL + SHIFT + ESC simultaneously to open Task Manager. Alternatively, you can right-click an empty area of the Taskbar and select **Task Manager**, or select it from the Quick Access menu. If the window appears compact, click "**More details**" (if visible) to show all tabs (Processes, Performance, App history, Startup apps, Users, Details, Services).



6. **End a Process (Settings App):** On the **Processes** tab, locate and right-click "**Settings**" (or "Windows Settings" if listed). Select **End Task**. This will close the Settings app. Note that for unsaved work in other applications, ending the task will result in data loss.

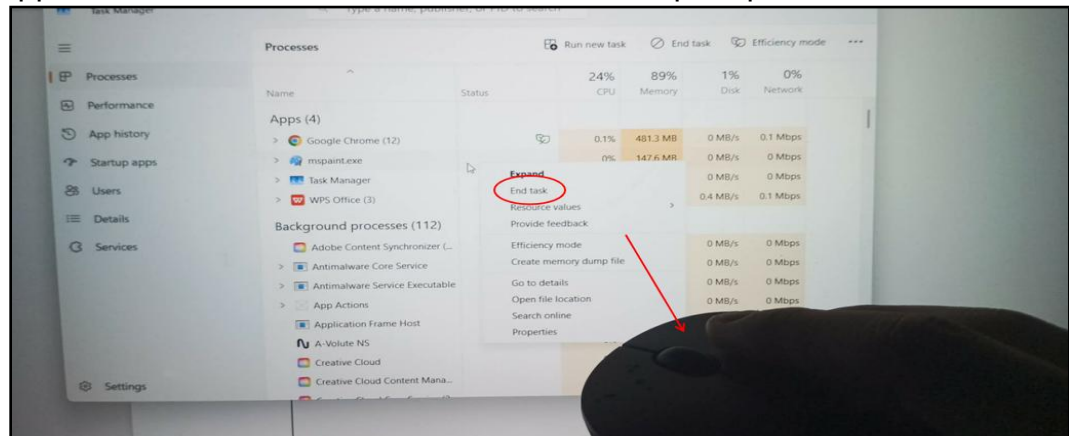


7. **Practice Ending a Task with Data Loss:**
- Launch **Paint** (you can search for it in the Start menu). Draw a simple picture in the Paint window but **do not save the file**.

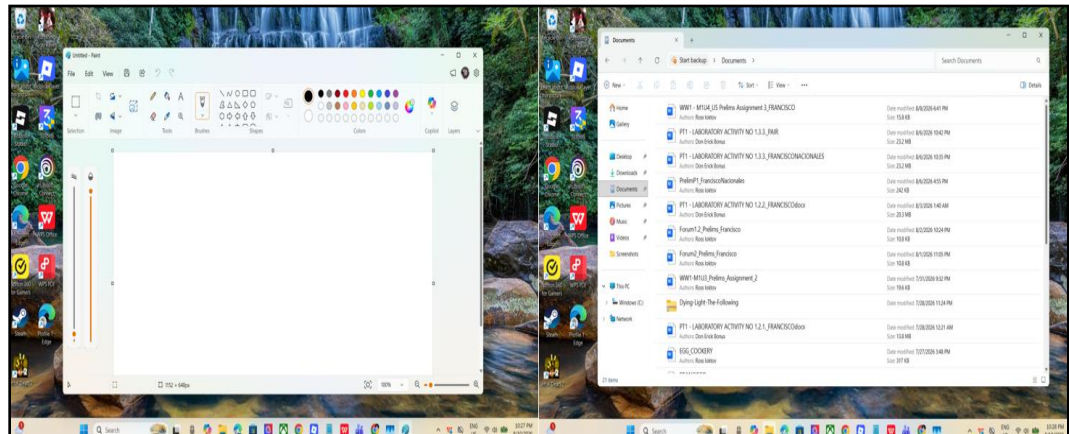




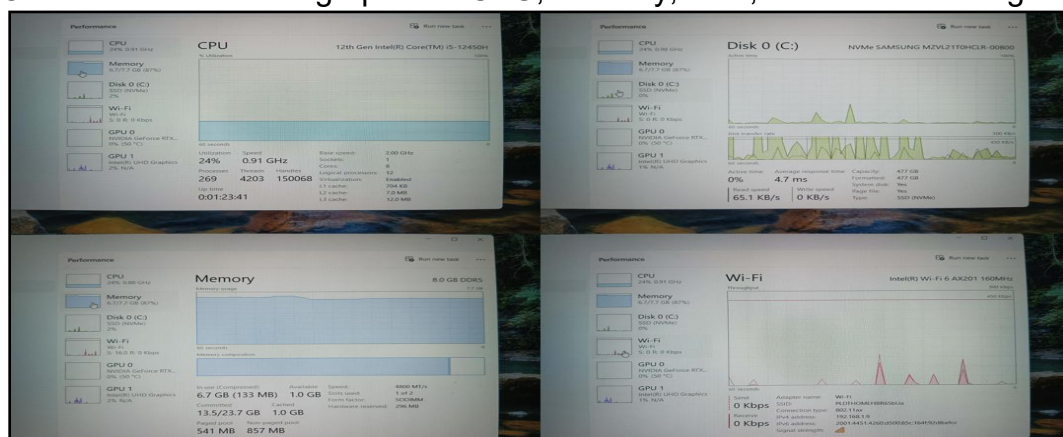
- b. Return to **Task Manager**. On the **Processes** tab, right-click the **"Paint"** application and select **End task**. Confirm when prompted.



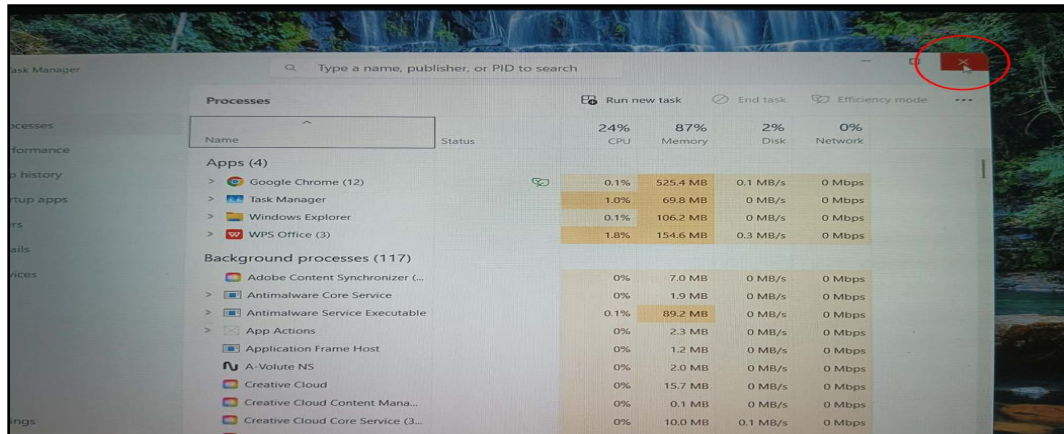
- c. Observe that your drawing is lost because it was not saved before the task was ended.



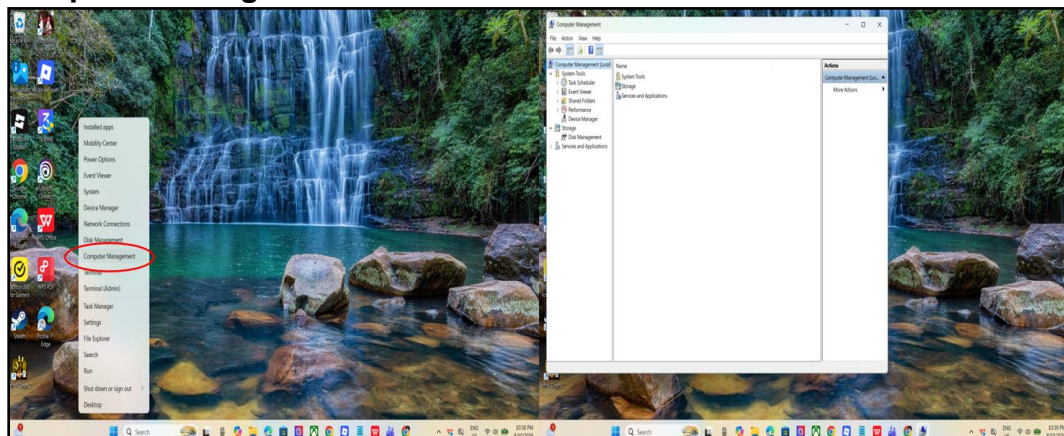
8. **Monitor Performance:** In Task Manager, click the **Performance** tab. Observe the real-time graphs for CPU, Memory, Disk, and Network usage.



9. **Close Task Manager.**

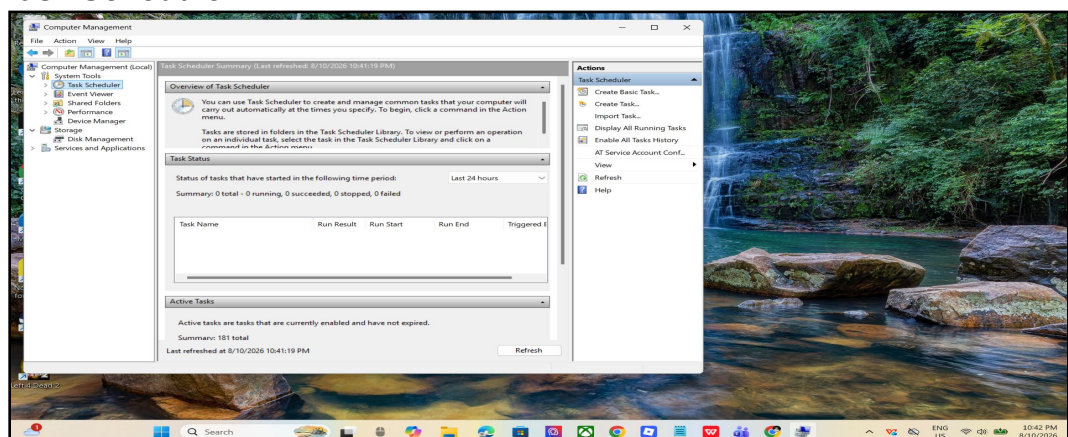


10. **Open Computer Management:** Right-click the **Start** button and select **Computer Management**.

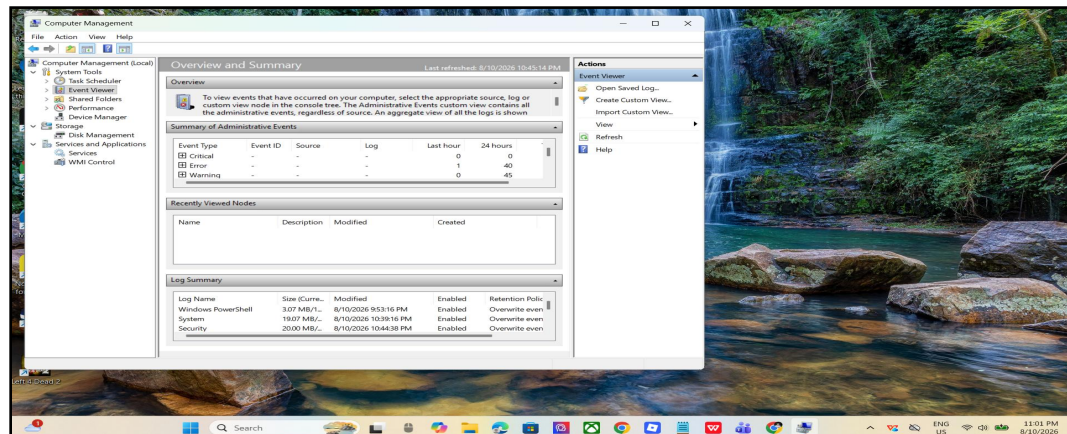


11. **Explore Management Snap-ins:** In the left-hand pane of the Computer Management console, explore the various administrative snap-ins available. These include:

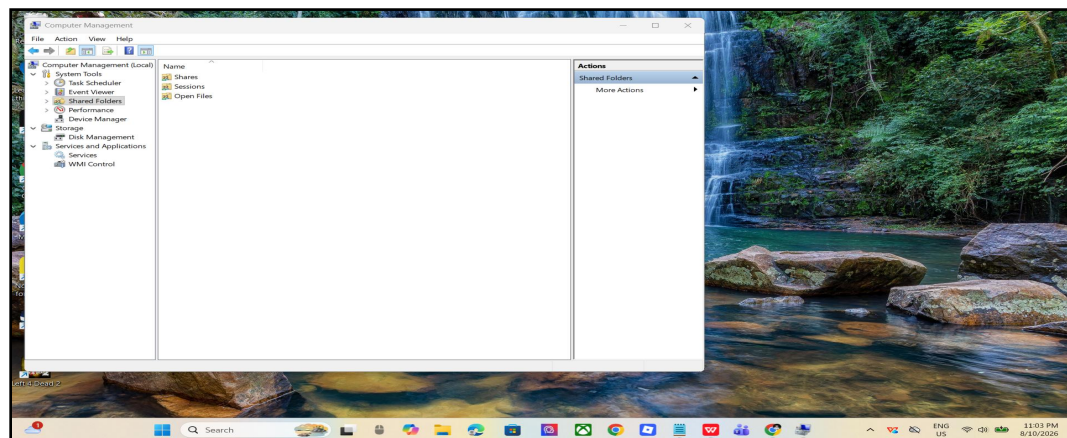
a. **Task Scheduler**



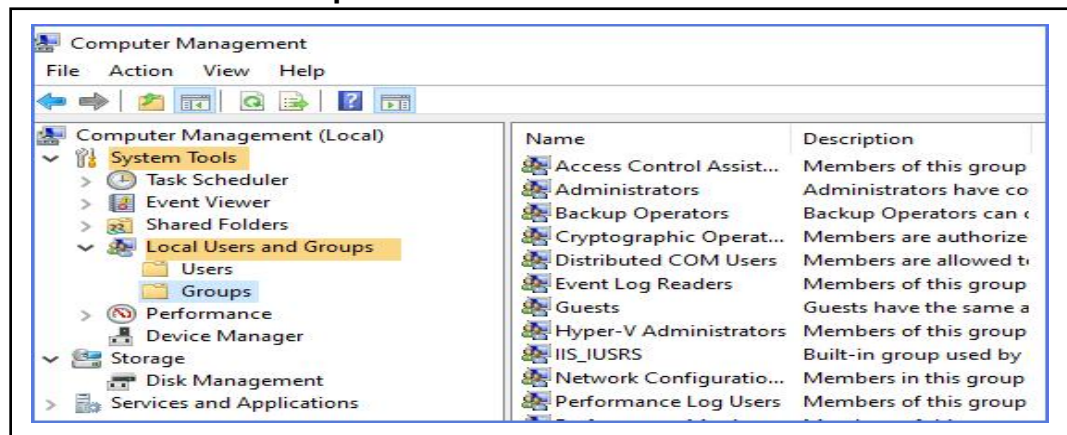
b. **Event Viewer**



c. Shared Folders

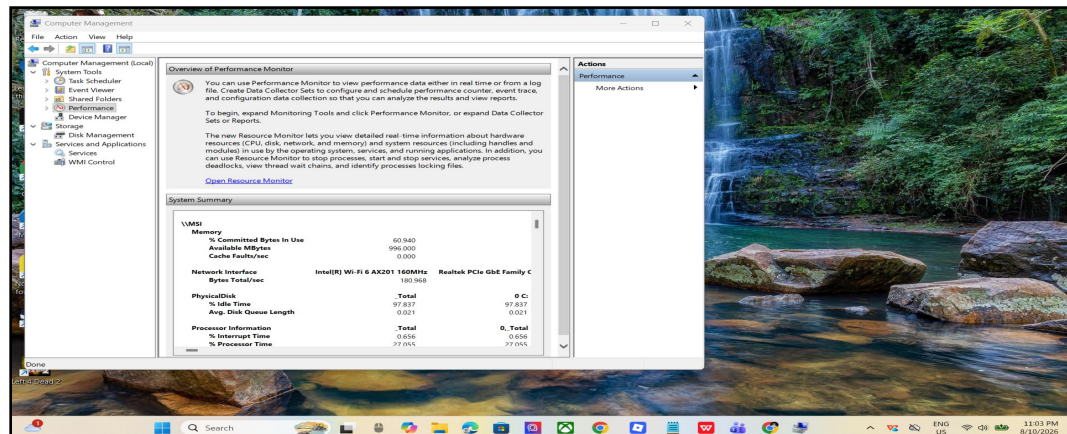


d. Local Users and Groups

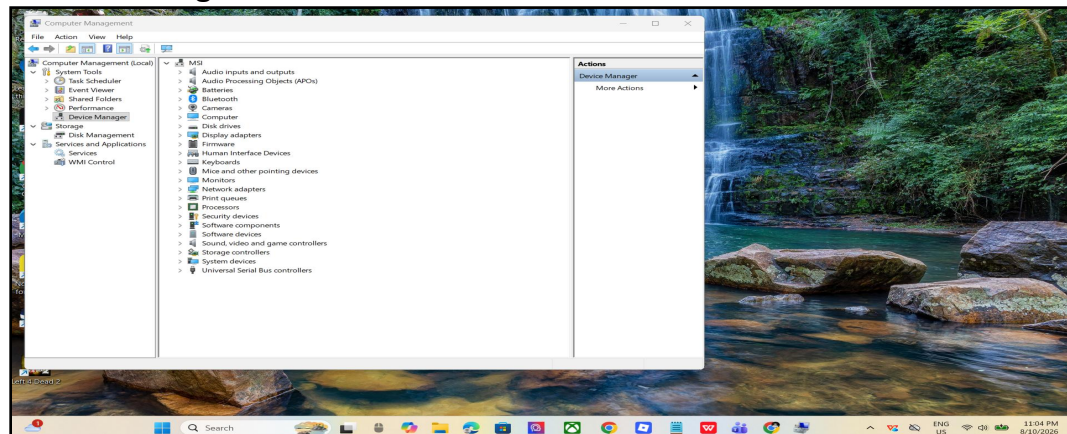


Source: <https://morgantechspace.com/2022/09/how-to-open-local-users-and-groups-console-on-windows-10-11.html>

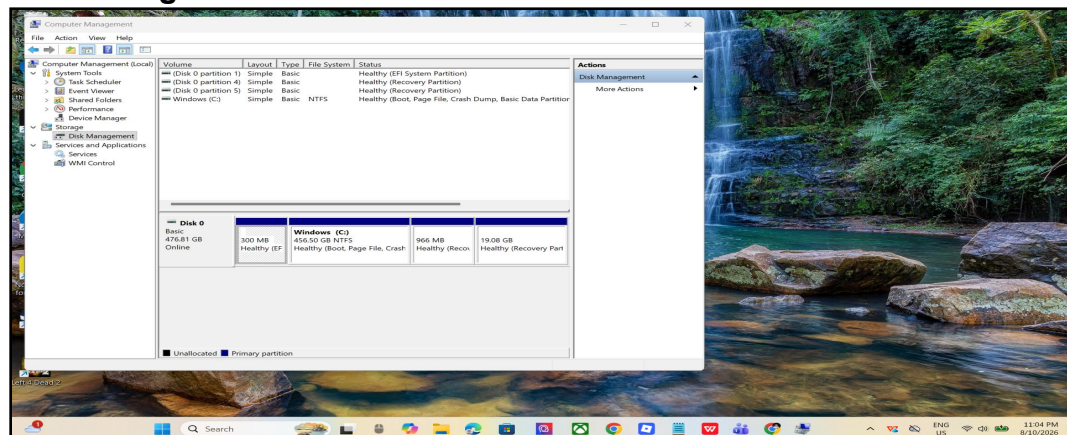
e. Performance Monitor



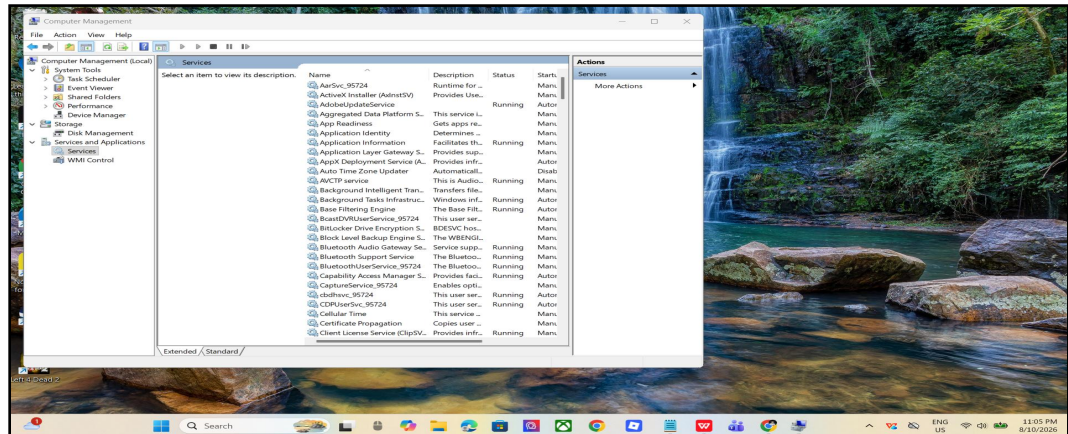
f. Device Manager



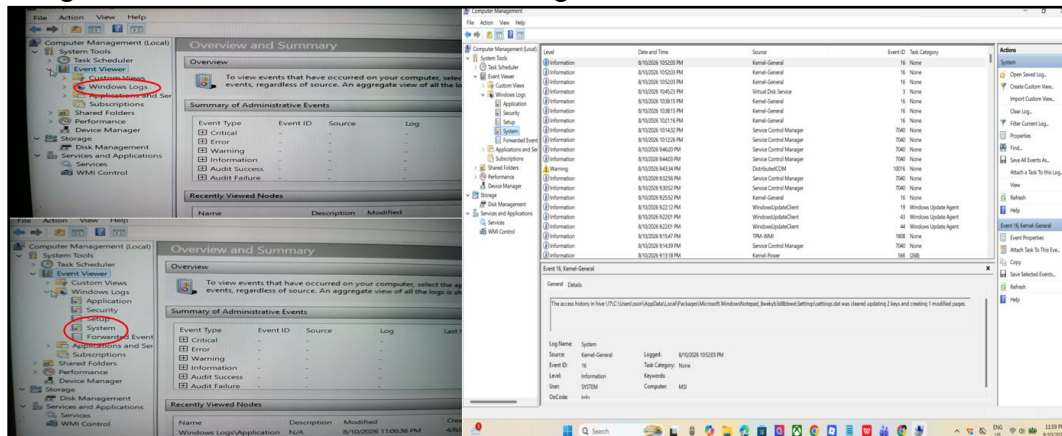
g. Disk Management



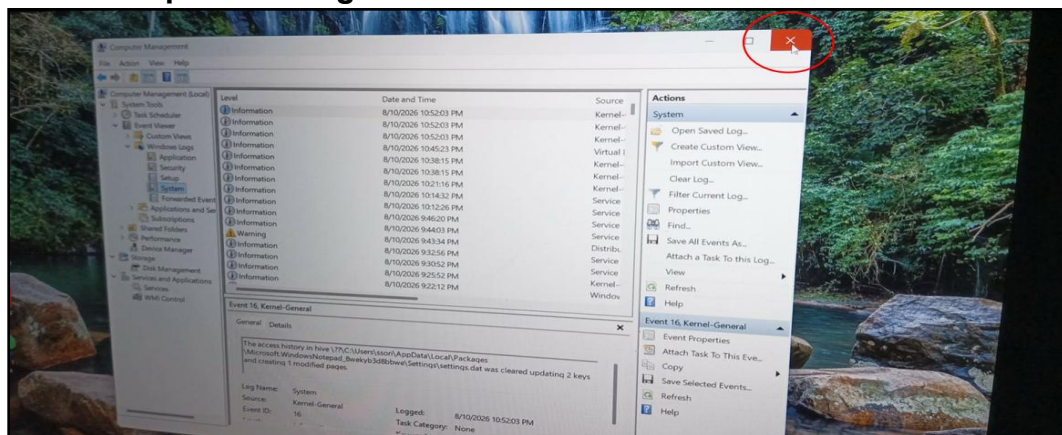
h. Services



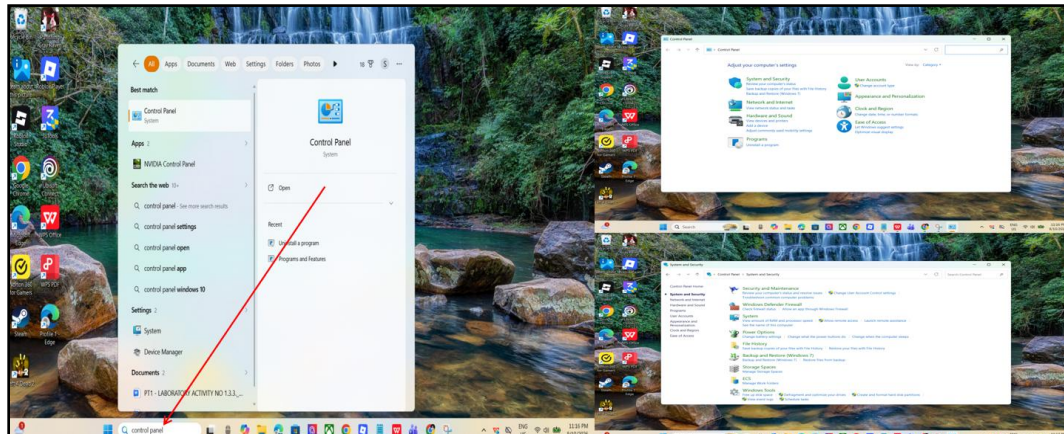
12. **Examine System Events:** Click **Event Viewer**, then expand **Windows Logs**, and click **System**. Examine some of the log entries, noting the different categories such as Information, Warning, and Error.



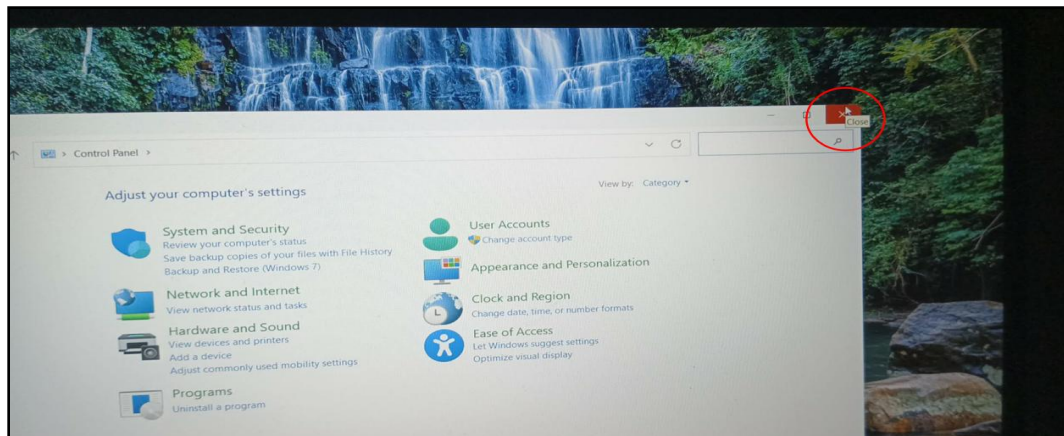
13. **Close Computer Management.**



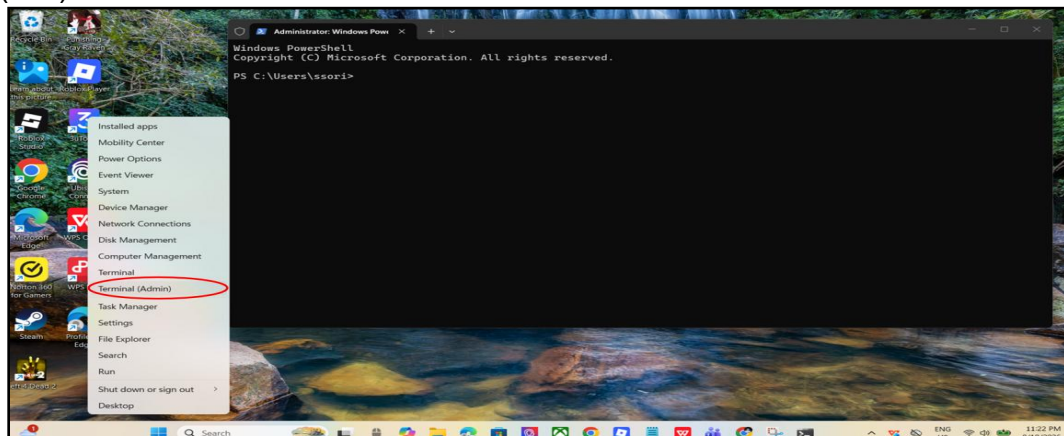
14. **Open Control Panel:** Click the **Start** button, then type control panel in the search bar. Click the **"Control Panel"** application icon that appears in the search results. Browse through the various applets (utilities) available.



15. Close Control Panel.



16. Launch Windows PowerShell: Right-click the **Start** button and select **Windows PowerShell** (or **Windows Terminal (Admin)** and then select PowerShell if prompted). This is an example of a Command Line Interface (CLI).

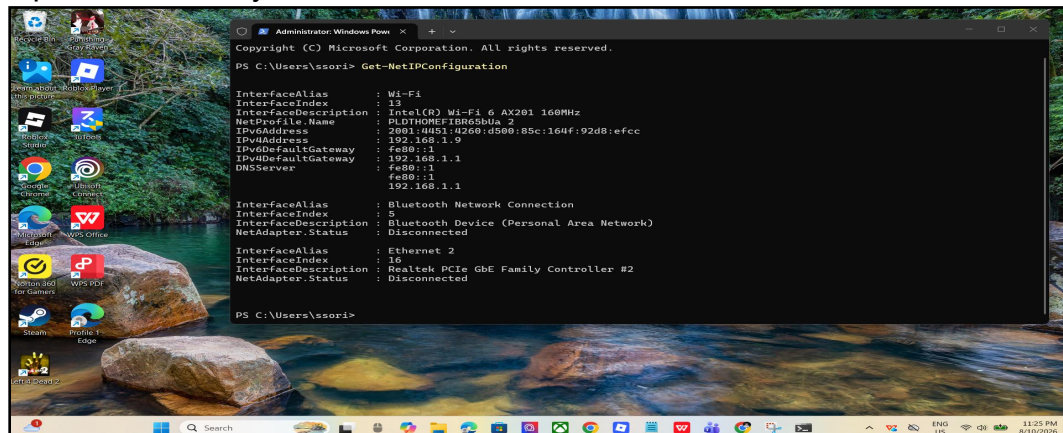


17. Execute a PowerShell Commandlet: In the PowerShell window, type the following command and press ENTER: `Get-NetIPConfiguration` This

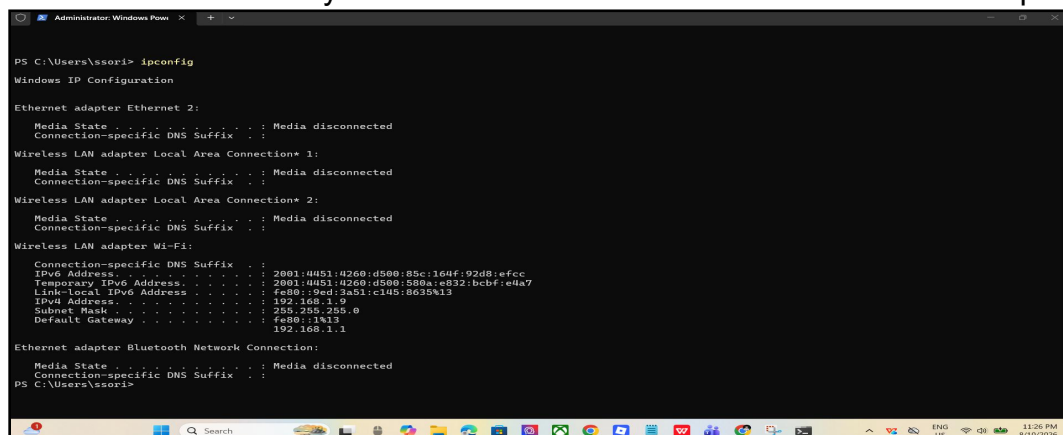


PowerShell commandlet displays detailed configuration information for your computer's network adapters.

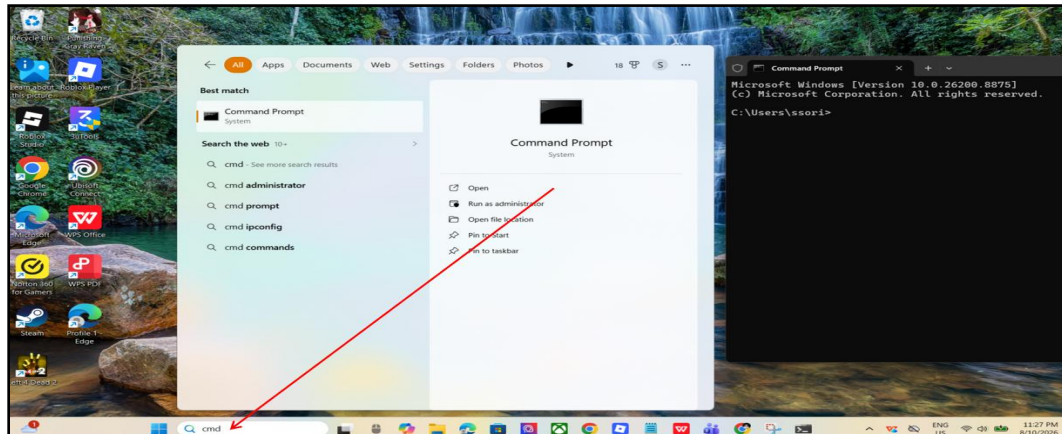
Note: PowerShell is typically case-insensitive, but using correct casing can improve readability.



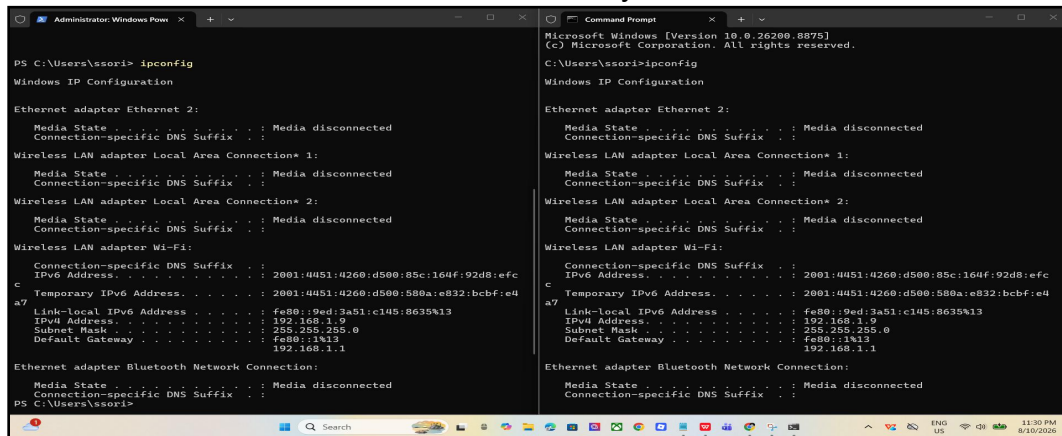
18. **Execute an Older Command in PowerShell:** Type the following command and press ENTER: `ipconfig` This older command also displays network configuration information. PowerShell is versatile and can run both its native commandlets and many commands from the traditional Command Prompt.



19. **Launch Command Prompt:** Click the **Start** button, then type `cmd` in the search bar. Click the "**Command Prompt**" application icon that appears in the search results.



20. **Execute a Command in Command Prompt:** In the Command Prompt window, type the following command and press ENTER: `ipconfig` Observe that this command executes in the same way as it did in PowerShell.



21. **Observe a Command Prompt Error:** In the Command Prompt window, type the following command and press ENTER: `Get-NetIPConfiguration` You will see an error message. This demonstrates that the Command Prompt environment **cannot understand (parse) PowerShell commandlets**. When working with CLIs, it's essential to know which commands are compatible with the specific environment you are using.



The screenshot shows two windows side-by-side. The left window is 'Administrator: Windows PowerShell' and the right is 'Command Prompt'. Both show the output of the 'Get-NetIPConfiguration' command. The PowerShell window shows detailed configuration for Wi-Fi, Bluetooth, and Ethernet adapters. The Command Prompt window shows a similar output but with a red circle around the error message: 'C:\Users\ssori>Get-NetIPConfiguration' 'Get-NetIPConfiguration' is not recognized as an internal or external command, operable program or batch file.

22. Close any open windows.

This screenshot is identical to the previous one, but with a red arrow pointing to the 'Close' button (the 'X' icon) in the title bar of the 'Command Prompt' window, indicating the instruction to close open windows.

V. REFLECTION:

Upon successful completion of "Lab 1.4.4: Using Management Interfaces," write a comprehensive reflection (between 150-250 words) addressing the following points:

- **Learning Outcomes:** What new knowledge or skills did you acquire or reinforce regarding the Windows user interface?
- **Challenges and Solutions:** Describe any difficulties or unexpected issues you encountered during the lab. How did you overcome them, or what steps would you take to resolve them in the future?
- **Real-world Relevance:** How do the skills and understanding gained in this lab relate to real-world scenarios, either in your personal life or a professional IT context?
- **Critical Thinking:** If you had to use an internet image for any step, explain why you couldn't perform the action physically and how finding the image contributed to your understanding.



Using Management Interfaces helped us strengthen our understanding of the Windows user interface and its management tools. We learned how different management interfaces can be used to view system information, manage users and settings, and perform administrative tasks. The activity also reinforced the importance of knowing where to find specific Windows tools instead of relying only on the graphical interface.

About the challenges, we only struggled a little with Computer Management because we were not very familiar with it. Still, we were able to accomplish the task by carefully following the instructions and exploring the available options.

The skills from this activity are relevant to real-world IT situations. Managing Windows settings and understanding administrative tools are useful when troubleshooting computers, configuring user accounts, maintaining systems, and assisting other users. These skills can also help us become more efficient when working with computers professionally.

For one of the steps in the activity, we used an image from the internet because the Local Users and Groups interface was not available on our Windows 11 Home computer. We used the image as a visual reference to understand what the interface should look like and where the required options were located. Although we could not perform that specific step directly, the image helped us understand the purpose and appearance of the management interface and why it may differ depending on the Windows edition.